FACULTY OF NATURAL SCIENCE
DEPARTMENT OF COMPUTER SCIENCE

CSE 2101: Software Engineering
System Testing: Student Records Management System (SRMS) Mobile Application

Client: Software Services & Educational Technology Applications (SSETA)
Date: December 23, 2025

Group 5
Kaylee Carter (1051799)
Khaleel Khan (1054193)
Ravindra Persaud (1052507)
Vimla Persaud (1052339)
Tulsidai Singh (1052682)

TABLE OF CONTENTS

1.0 Introduction

The System Testing phase is a critical component of the SDLC. It is the process of checking that a system complies with its stipulated requirements. The purpose of system testing is to validate the complete and integrated system to ensure that all functional requirements are working correctly from an end-user perspective.

This document provides a set of detailed system test cases for the Mobile Application of the Student Records Management System. The test cases have been designed to test major system functionalities, including student registration, managing payments, viewing academic records, timetable consistency, and application processing. Each test case will be carried out based on the standard testing template provided, containing distinctly defined pre-conditions, test steps, test data, expected results, and post-conditions.

These test cases will help ensure that the SRMS Mobile Application is reliable, enforces academic and financial policy appropriately, and ensures security and consistency in the user experience for students. The tests also contribute to the detection of defects at an early stage and verification that the system meets the institutional requirements before its deployment.

2.0 Test Case One

2.1 Course Registration - Prerequisite Validation CR-3

Test case ID	TC_CR_01
Test priority (Low/Medium/High):	High
Test Title/Name:	Validates prerequisite enforcement during course registration
Test Summary/Description	To ensure that the SRMS Mobile Application validates course requirements during the registration process and stops students from selecting courses when they haven’t met the prerequisites, showing an appropriate error message.

Pre-conditions	**1.** The student is logged into the SRMS using a valid USI and password. **2.** The course registration session is currently open. **3.** The student has an academic record which is stored in the SRMS database. **4.** The course selected has at least one prerequisite which is defined in the SRMS system. **5.** The student has not completed the required prerequisite course. **6.** The SRMS unified API is available and functioning.
Dependencies	**1.** The course registration functional requirements (CR-3 and CR-4) must be implemented. **2.** The student academic records must be available for prerequisites validation. **3.** SRMS unified API must be active to verify the course selection.
Test Steps	**1.** Launch the SRMS mobile application on the mobile device (Android or iOS test device). **2.** Log in using the provided student test account with a valid USI (Student USI: *1234567)* and password . **3.** Once logged in, from the dashboard select the registration to access the registration module and start the session. **4.** Verify that the system displays the list of available courses for the student's programme and upcoming semester. **5.** Locate the course CSE2100 - Data Structures and Analysis of Algorithms. **6.** Select the CSE2100 to view its course details. **7.** Tap Add/ Register to attempt to add the course. **8.** Allow the system to perform the validation. **9.** Observe the system response message displayed on the screen. **10.** Navigate to Selected Courses / Provisional Registration List. **11.** Confirm that CSE2100 does not appear in the selected course list.

	12. Attempt to proceed to Preview and Submit and confirm the course is not included.
Test Data	**1.** Student Test Account ● USI: *1234567* ● Programme: Associate of Science (Computer Science) ● Residency Status: Guyanese ● Academic Status: Active **2.** Completed courses ● CSE1200 - Introduction to Computing **3.** Not Completed Courses ● CSE1201 - Introduction to Programming **4.** Target Course for Registration ● CSE2100 - Data Structures and Analysis of Algorithms **5.** Prerequisite Requirement ● CSE1201 - Introduction to Programming (Must obtain a pass before registering for CSE2100)
Expected Results	1. When the student wants to add CSE2100 - Data Structures and Analysis of Algorithms, the system runs the validation using the student's academic record stored in the SRMS. 2. Since the student has not finished CSE1201 - Introduction to Programming the system rejects the course selection. 3. The system then displays a clear error message on screen such as "Prerequisite not satisfied: CSE1201 - Introduction to Programming is required for CSE2100." 4. The course CSE2100 is not added to the selected courses / Provisional Registration list. 5. On the Preview and Submit screen, CSE2100 is not displayed in the registration summary and not included in the overall credit and course total. 6. No registration record which violates course requirements is saved or submitted to the SRMS backend for that course. 7. The system also accommodates special cases which may be reviewed and adjusted by the Head of Department

	(HOD) during the review and approval process, where applicable.
Post Condition	1. The student's registration session remains active and unchanged. 2. CSE2100 - Data Structures and Analysis of Algorithms is not included in the students registered or provisional course list. 3. No registration record which violates course requirements is stored in the SRMS database. 4. The student may continue the registration by selecting eligible courses or proceed with registration for courses that meet all prerequisites requirements.

3.0 Test Case Two

3.1 Payment Management - Financial Status Block PM-6/APR-2

Test case ID	TC_PM_01
Test priority (Low/Medium/High):	High
Test Title/Name:	Verify academic record and registration lockout for foreign students with outstanding balances and automatic unblock after successful payment
Test Summary/Description	This test case verifies that the SRMS correctly blocks access to Academic Records and Course Registration for foreign students who have an outstanding financial balance. It also confirms that once the outstanding balance is cleared using any supported payment method, the system automatically updates the student's

	financial status and immediately restores access to academic services without requiring manual intervention.
Pre-conditions	1. The student is logged into the SRMS Mobile Application as a Foreign Student (Citizenship: St. Lucia). 2. The student's financial status in the SRMS database is marked as "Uncleared", with an outstanding balance greater than $0. 3. The academic registration period is currently active. 4. At least one supported payment option is available in the system. 5. The student has an existing academic record stored in SRMS.
Dependencies	1. FR-PM-6: The system shall block academic records and registration access for uncleared foreign students. 2. APR-2: Academic records access must reflect real-time financial clearance status. 3. FR-PM-3: The system shall automatically update financial status upon successful payment. 4. Financial, academic, and registration modules must remain synchronized.
Test Steps	1. From the dashboard, tap the Academic Records icon. 2. Observe the system response displayed on screen. 3. Return to the dashboard and navigate to the Finances module. 4. Locate the outstanding invoice with a balance of $45,000 GYD. 5. Select the invoice and choose Pay using any available payment option provided by the system. 6. Complete the payment process using valid test credentials. 7. Wait for the system to display a payment confirmation message. 8. Return to the dashboard and tap Academic Records again. 9. Attempt to access Course Registration.

<table>
<tr><td>Test Data</td><td>Student Test Account
Student ID: 1054193
Citizenship Status: Foreign (St. Lucia)
Programme: Bachelor of Information Technology
Academic Status: Active
Financial Information
Financial Status (Before Payment): Uncleared
Outstanding Balance: 45,000 GYD
Invoice ID: INV_2025_118
Payment Method: Any system-supported payment option</td></tr>
<tr><td>Expected Results</td><td>Step 2:
The system displays a blocking alert such as:
“Access Denied: Outstanding balance of $45,000 GYD. Please clear fees to proceed.”
Academic Records and Course Registration remain inaccessible.
Step 7:
The system confirms successful payment with a message such as:
“Payment Successful. Financial status updated.”
Steps 8-9:
Academic Records load successfully and display the student’s grade list.
Course Registration becomes accessible without restriction.
No logout, refresh, or manual approval is required.</td></tr>
<tr><td>Post Condition</td><td>The student’s financial status is updated to “Cleared”.
The outstanding invoice balance is updated to $0.
All academic and registration restrictions are removed.</td></tr>
</table>

	• The SRMS database remains synchronized across all relevant modules.

4.0 Test Case Three

4.1 Student Application - Document Upload block SA-7/8

Test case ID	TC_SA_01
Test priority (Low/Medium/High):	High
Test Title/Name:	Verify the upload of valid supporting documents for the University of Guyana Student application.
Test Summary/Description	This test case verifies that a prospective student can successfully upload all required supporting documents, such as National ID/Passport and academic certificates, during the Student Application process in the SRMS Mobile Application. It ensures that uploaded documents are validated, stored, and correctly linked to the applicant's application record in the SRMS database that is accessed via the API.
Pre-conditions	1. The SRMS Mobile Application is installed on a supported mobile device. 2. The prospective student has an existing applicant account or has successfully created one. 3. The user is logged into the mobile application. 4. The Personal Information, such as Home Address, phone number, parent (s) personal information, and so on, has been filled out and saved. 5. The user has selected a programme of study.

	6. The SRMS application is connected to the SRMS backend via the API. 7. The device has granted permission for the SRMS application to access local storage. 8. A stable internet connection is available.
Dependencies	1. The Document Upload block can only be accessed after earlier application sections are completed. 2. Any failure to save personal, address, education, or program selection data will prevent this test from executing. 3. “FR-SA-7: The system must support uploading the required supporting documents.” Additionally, “FR-SA-8: The system must validate all the necessary documents before allowing the user to proceed.” If these requirements are not fully implemented, document upload validation cannot be tested accurately. 4. NFR-Security: Restrict file types to prevent malware (PDF/Image only). Any other file format should be rejected and show a user-friendly error message to the user. 5. The system shall also be configured to accept files at a maximum file size of 5MB and display a user-friendly error message if it exceeds the defined file size of 5MB.
Test Steps	1. Open the SRMS Mobile Application. 2. The prospective student will log in to their student account in the SRMS mobile application. 3. Select “Apply to UG” from the main menu. 4. Complete all required sections until the Document Upload screen is displayed. 5. Tap “Upload Document” for a required document. 6. Select the respective file from the device's local storage. 7. Confirm the selected file. 8. Repeat the process for all required documents. 9. Verify all documents have a successful upload status. 10. Go to the Application Summary screen.

Test Data	Test Data 1 ● Document Type: National ID ● File Name:kaylee_id.pdf ● File Format: PDF ● File Size: 450KB Test Data 2 ● Document Type:CSEC Certificate ● File Name:kaylee_csec_certificate.jpg ● File Format:JPEG ● File Size: 5MB Test Data 3 ● Document Type: Birth Certificate ● File Name: kaylee_carter_birth_certificate.docx ● File Format: .docx ● File Size:1MB
Expected Results	Expected Results for Test Data 1 ● When the applicant uploads the National ID file kaylee_id.pdf, the system accepts it and successfully uploads the document because the PDF format and 450 KB file size are within the allowed limits. Expected Results for Test Data 2 ● The CSEC Certificate file, kaylee_csec_certificate.jpg is also accepted and uploaded successfully, as the JPEG format is supported and the 5 MB file size meets the maximum allowed limit. Expected Results for Test Data 3 ● However, when the applicant attempts to upload the Birth Certificate file kaylee_carter_birth_certificate.docx, the system rejects the file because the .docx format is not

	allowed. It then displays an error message such as "File format not supported..Upload PDF, JPEG or PNG", and does not allow the user to the Preview and Submit screen until a valid document is uploaded.
Post Condition	1. The system allows successful uploads only for documents that meet both file format and file size validation rules. 2. The system is configured to have documents completion compulsory and denies submission when any required document fails validation. 3. The application remains in "In Progress" status until all required documents are successfully uploaded.

5.0 Test Case Four

5.1 Student Application - Block Duplicate Applications UC-02 Exception 9a

Test case ID	TC_SA_02
Test priority (Low/Medium/High):	Medium
Test Title/Name:	Verify that the system prevents the submission of duplicate applications for the same academic year.
Test Summary/Description	We need to confirm that if a student has already submitted an application that is "Under Review" or "Accepted", the system stops them from submitting a second one to prevent confusion.

Pre-conditions	1. User is logged in as a Prospective Student. 2. The user has already submitted one application for the current academic year (2025/2026). 3. The status of that previous application is “Submitted.”
Dependencies	UC-02 Exception 9a: The system detects an existing application and redirects an existing application. FR-SA-11: Send complete application to backend (which performs the check)
Test Steps	1. Log in with the test account. 2. Tap the “Start New Application” button on the home screen. 3. Select a programme (eg. Bachelor of Computer Science) and fill in the required Personal Information fields. 4. Skip to the final step and tap “Submit Application.”
Test Data	Application ID: App_2025_99 Existing Record: Application for Associate Degree in Chemistry (Status: Submitted). New Attempt: Application for Bachelor of Computer Science
Expected Results	The system should display a notification/alert immediately: “Duplicate Application Detected. You have already submitted an application for this academic year. Please check ‘My Status’ for updates.” The app should not send a new record to the database.
Post Condition	The user is redirected to the Application Status screen. The total number of applications for this user in the database remains at 1.

6.0 Test Case Five

6.1 Timetable Viewing - Timetable Consistency when course is dropped CR-12/TV-4

Test case ID	TC_TV_01
Test priority (Low/Medium/High):	High
Test Title/Name:	Verify timetable consistency when a registered course is dropped or added
Test Summary/Description	This test case will verify that the SRMS Mobile app is functioning properly to update the student's schedule when a course is added or dropped within the registration period. The purpose is to validate that the schedule information is up-to-date, consistent, and in sync with the courses being registered by the student at any given time to avoid any inconsistencies such as schedule conflicts, absent classes or out-of-date data.
Pre-conditions	1. The student logs into the SRMS Mobile Application using his/her valid USI and password. 2. The course registration period is ongoing. 3. The student has an active academic status. 4. The student is already registered for at least two courses that appear on the timetable. 5. In SRMS, the timetable information for each course (day, time, venue, and lecturer) is predefined and is stored in the SRMS database. 6. The timetable viewing module (TV-4) is fully functional. 7. The unified API on the SRMS system is working properly.
Dependencies	1. The Course Add/Drop feature should be functional. 2. Timetable viewing functionality must retrieve real-time data.

	3. Student registration data has to be in synchronization with the timetable data. 4. The backend database also has to maintain timetable slots for all courses registered.

Test Steps	1. Start the SRMS Mobile app on the test mobile device (Android/IOS).
	2. You will need to use a student test login (USI: 1234567).
	3. From the dashboard, navigate to My Timetable.
	4. Note the current list of timetable entries.
	5. Verify that the schedule reflects the following courses:
	    ● CSE2101 - Software Engineering
	    ● CSE2102 - Database Systems Design and Information Management
	6. Return to the dashboard and choose the option of Course Registration.
	7. Find CSE2102 - Database Systems Design and Information Management in the list of registered courses.
	8. Tap Drop Course and confirm the action when prompted.
	9. Wait for the system to process the request and display confirmation.
	10. Go back to My Timetable.
	11. Confirm whether CSE2102 is no longer in the timetable.
	12. Go back to Course Registration.
	13. Find ISY2100 - Information Systems Analysis in the list of courses that are available.
	14. Tap Add/Register Course.
	15. Let the system finish the validation and registration process.
	16. Go back to My Timetable.
	17. Verify that ISY2100 now appears with the correct day, time, venue and lecturer.

Test Data	Student Test Account ● USI: 1234567 ● Programme: Associate of Science (Information Technology) ● Academic Status: Active

	• Semester: Year 2, Semester 1 Initial Registered Courses • CSE2101 - Software Engineering, Mon 9-11 am • CSE2102 - Database Systems Design and Information Management, Wed 1-3 pm Dropped Course • CSE2102 - Database Systems Design and Information Management Added Course • ISY2100 - Information Systems Analysis • Scheduled Day/Time: Fri 10 AM - 12 PM • Venue: CSI Lab • Lecturer: Ms Sandra Khan

Expected Results	1. When the student un-enrolls from CSE2102, the system will delete this course from his/her registration records.
	2. The revised schedule will be automatically generated or when re-entered, and CSE2102 will not be shown in it
	3. There is no empty or broken slot on the timetable after the course has been dropped.
	4. When ISY2100 is added, the system saves the registration successfully.
	5. The timetable updates immediately to include ISY2100 with:
	  • Correct day
	  • Correct time
	  • Correct venue
	  • Correct lecturer
	6. The timetable shows only registered classes that are up to date and do not repeat.
	7. There are no error messages or discrepancies displayed in the process of the update.

Post Condition	1. The student's timetable accurately matches the student's final registered course list. 2. Courses that have been dropped no longer appear in timetable records. 3. Newly added courses are fully shown in the timetable view. 4. The SRMS database remains synchronized across registration and timetable modules. 5. The student can confidently rely on the timetable for academic planning.

Collaborative Summary

Division of Labour:

Name	Tasks	Signature
Ravindra Persaud	Test Case 1	R. Persaud
Khaleel Khan	Test Case 2	
Kaylee Carter	Test Case 3	K. Carter
Tulsidai Singh	Test Case 4	T. Singh
Vimla Persaud	Test Case 5	Vimla Persaud

Collaboration Strategies and General Experiences:

Our group used a shared Google Doc for real-time collaborative writing and a WhatsApp group for daily communication and coordination. This strategy worked well, as the shared document allowed everyone to see progress and contribute easily. Regular check-in meetings were held via Google Meet to ensure everyone was aligned.

Challenges and Resolutions:

Our main challenge was ensuring a consistent writing style and tone across the document, since each section was written by a different person. We handled this by designating one member as the final compiler, responsible for reviewing the entire document and making minor edits to ensure it read as a single, cohesive report.